LECTURE NOTES ON HYPERBOLIC DYNAMICS

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1. Introduction

In this course we shall deal with *autonomous deterministic dynamical systems.* They can be used to represent or modelize any phenomenon that changes along the time in such a way that the current state of the system completely determines the future evolution and the law that drives this evolution does not change along the time. Mathematically, an **autonomous deterministic dynamical system** (a **dynamical system**, for short) is given by a set M, usually called the **phase space,** which represents the set of possible states of the system, a subsemigroup $\mathbb{T} \subset \mathbb{R}$ that represents the **time set,** and *"a law"* $\Phi\colon \mathbb{T} \times M \to M$ that determines the evolution of the system in such a way that given the current state of our system $x_0 \in M$, at time $t \in \mathbb{T}$ the corresponding state will be $x_t = \Phi(x_0, t)$. The autonomy of the system is reflected in the so called **flow** (or **semi-flow**) condition function Φ satisfies:

$$\Phi(t+s,x) = \Phi\big(t, \Phi(s,x)\big) = \Phi\big(t, \Phi(s,x)\big), \quad \forall t,s \in \mathbb{T},\ \forall x \in M. \tag{1}$$

We will consider two different kinds of dynamical systems: **discrete time** and **continuous time** ones, i.e. where $\mathbb{T} = \mathbb{N}_0$ and $\mathbb{T} = \mathbb{R}_{\geqslant 0}$, respectively. Moreover, we will also consider **invertible systems** where the current state of the systems does not just determine the future but the past as well. When dealing with invertible discrete dynamical systems, the time set will be $\mathbb{T} = \mathbb{Z}$, and in the case of invertible continuous time ones, we will have $\mathbb{T} = \mathbb{R}$.

The main goal of Dynamical Systems consists in describing the **orbits of the systems,** i.e. the sets

$$\mathcal{O}_\Phi(x) := \{\Phi(t,x) : t \in \mathbb{T}\}, \quad \forall x \in M.$$

Exercise 1.1 (Discrete time generator)**.** Given a discrete time dynamical system $\Phi\colon \mathbb{T} \times M \to M$, show that Φ satisfies the flow condition (1) if and only if there is $f\colon M \circlearrowleft$ such that

$$\Phi(n,x) = f^n(x), \quad \forall x \in M,\ \forall n \in \mathbb{N}_0,$$

where the iteration of f is inductively defined by $f^0 := id_M$ and $f^{n+1} = f \circ f^n$, for every $n \geqslant 0$. Show that Φ is invertible if and only if f is bijective and, in such a case, one can define $f^{-n} := \big(f^{-1}\big)^n$, for all $n \geqslant 0$.

On the other hand, one can find *infinitesimal generators* for continuous time smooth dynamical systems:

Exercise 1.2 (Continuous time generator)**.** Let M be a differentiable closed manifold (i.e. compact and boundaryless) and $\Phi\colon \mathbb{R} \times M \to M$ a continuous time dynamical system (that satisfies the flow condition 1). If we assume Φ is a C^2 map and we define $X(p) := \partial_t \Phi(t,p)\big|_{t=0} \in T_pM$, for every $p \in M$, then the function $\Phi(\cdot, x_0)\colon \mathbb{R} \to M$ is the unique solution of the *initial value problem*

$$\dot{x}(t) = X\big(x(t)\big), \quad x(0) = x_0,$$

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for every $x_0 \in M$. In other words, the vector field X is the infinitesimal generator of the flow Φ.

As previously mentioned, the primary objective of Dynamical Systems is to understand the orbits of various systems, particularly their asymptotic behavior as time goes to infinity. From a historical perspective, there are three pivotal moments in the development of this mathematical theory. The first can be attributed to Isaac Newton, who laid the groundwork for Calculus and employed its principles to solve differential equations. During this era, the approach to studying a specific continuous-time dynamical system involved solving the corresponding differential equation, followed by an analysis of the obtained solutions.

The so-called *"restricted two-body problem"* is a remarkable question that was answered at that time:

Problem 1.1. The *"restricted two-body problem"* refers to a simplified model used to analyze the motion of two celestial bodies under the influence of their mutual gravitational attraction, where one body is significantly more massive than the other. In this framework, the more massive body is assumed to be stationary, while the second body is treated as a point mass that moves in the gravitational field created by the first one. This approach allows for the derivation of equations of motion for the second body without needing to account for its gravitational effects on the primary, simplifying the analysis. This model is particularly useful in celestial mechanics, where it can be applied to scenarios such as a satellite orbiting a planet or a comet passing by a star.

More precisely, the *restricted two-body problem* consider in describing the orbit of massive point p, whose location in the space at time $t \in \mathbb{R}$ is given by $x(t) \in \mathbb{R}^3$, moving in a central field force in $\mathbb{R}^3$ that is generated by a massive point of mass $M > 0$ which is fixed at the origin. According to Newton's laws, the trajectory $x(t) \in \mathbb{R}^3$ is the solution of the following initial value problem:

$$\begin{cases} \ddot{x}(t) = -\dfrac{GM}{\|x\|^3} x(t), \\ x(0) = x_0, \\ \dot{x}(0) = v_0, \end{cases} \tag{2}$$

where $G > 0$ is the universal constant of gravitation, $x_0 \in \mathbb{R}^3$ and $v_0 \in \mathbb{R}^3$ are the initial position and velocity, respectively.

Using classical *quadrature techniques* one can show that the solutions $x(t)$ of (2) are either straight lines through the origin (when x_0 and v_0 are collinear vectors in $\mathbb{R}^3$), or quadratic curves (i.e. ellipses, parabolas or hyperbolas).

The second pivotal moment in the development of Dynamical Systems can be attributed to Henri Poincaré. At the end of XIXth century it was already pretty clear that there were many ordinary differential equations that could not be solved with quadrature methods. A very simple and well-known example is the following one:

Example 1.1. There is no *elementary (smooth) function* $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f'(x) = e^{-x^2}, \quad \forall x \in \mathbb{R}.$$

Another example, which was fundamental in development of so called *qualitative theory of differential equations* is the 3-*body problem:*

Problem 1.2 (n-body problem)**.** The *n-body problem* refers to the challenge of predicting the individual motions of a group of n celestial objects that interact with each other through gravitational forces. The complexity arises from the fact that

each body exerts a gravitational force on every other body, leading to a system of coupled differential equations. If $x_i(t) \in \mathbb{R}^3$ represents the position of the i-th object at time $t \in \mathbb{R}$ and $m_i > 0$ denotes its mass, then Newton's second law determines that these quantities are related by the following system of ordinary differential equations

$$\begin{cases} \ddot{x}_i(t) = \sum_{j \neq i} G \dfrac{m_j}{\|x_i(t) - x_j(t)\|^3} \big(x_i(t) - x_j(t)\big), \\ x_i(0) = x_{i,0}, \\ \dot{x}_i(0) = v_{i,0}, \end{cases} \tag{3}$$

for $1 \leqslant i \leqslant n$, and where $x_{i,0}, v_{i,0} \in \mathbb{R}^3$ denotes the position and the velocity of the i-th object at time $t = 0$, respectively.

It is pretty remarkable that, in general, the system of ODEs given by (3) cannot be solved by "quadrature methods" when $n \geqslant 3$. In fact, their solutions are not elementary functions in general.

The n-body problem, as outlined in Problem 1.2, served as a primary motivation for Henri Poincaré in developing the Geometric Theory of Differential Equations. Poincaré posited that if explicit solutions to a given differential equation cannot be readily obtained, one should at least strive to describe some interesting geometric properties of those solutions.

Indeed, many questions regarding the solutions of differential equations are more qualitative than quantitative, or more geometric than analytic. For instance, a fundamental qualitative inquiry concerning the system of ordinary differential equations in (3) is whether every solution is periodic. In some cases, even when solutions can be expressed using elementary functions, addressing certain qualitative questions based solely on the explicit formulas can prove to be quite challenging.

The third pivotal moment in the evolution of Dynamical Systems can be attributed to the schools of Steve Smale in the US and Aleksandr Andronov in the USSR. At this juncture, it became evident that the geometric properties of orbits, trajectories, or solutions to ODEs could change dramatically with variations in system parameters. For example, in Problem 1.2, there are $3n$ parameters—comprising the n masses, n initial positions, and n initial velocities. In certain instances, even minor changes to these parameters can lead to completely different geometric properties for the respective solutions. Thus, it becomes crucial to identify geometric properties of orbits that are robust or persistent under small perturbations of system parameters.

These three historical stages in the development of dynamical systems are cleverly (and provocatively) summarized by Étienne Ghys in his laudation for Artur Avila's Fields Medal: In Newton's time, you were given an ODE and tasked with finding its solutions. In Poincaré's era, you were given an ODE and asked to elucidate interesting geometric properties of its solutions. In Smale's time, you were given no ODE at all and challenged to say something interesting!

2. Some examples of Dynamical Systems results

A very first example of a result in Dynamical Systems might be the following classical

Theorem 2.1 (Banach fixed point theorem)**.** *Let (M, d) be a complete metric space and $f\colon M \circlearrowleft$ be* ***uniform contraction,*** *i.e. f is a Lipschitz function and*

$$\lambda = \operatorname{Lip}(f) := \sup_{x \neq y} \frac{d\big(f(x), f(y)\big)}{d(x, y)} < 1.$$

Then there exists a unique point $p \in M$ such that $f(p) = p$ and it holds

$$\lim_{n\to+\infty} f^n(x) = p, \quad \forall x \in M.$$

Proof. Given any $x \in M$, one can show by induction on n that

$$d\big(f^n(x), f^{n+1}(x)\big) \leqslant \lambda^n d(x, f(x)), \quad \forall n \in \mathbb{N}_0.$$

So, for every $m, n \in \mathbb{N}$ it holds

$$\begin{aligned} d\big(f^m(x), f^{m+n}(x)\big) &\leqslant \sum_{i=0}^{n-1} d\big(f^{m+i}(x), f^{m+i+1}(x)\big) \\ &\leqslant \sum_{i=0}^{n-1} \lambda^{m+i} d\big(x, f(x)\big) \leqslant \frac{\lambda^m}{1-\lambda} d\big(x, f(x)\big). \end{aligned}$$

Hence, the sequence $\big(f^n(x)\big)_{n\geqslant 0}$ is Cauchy, and since M is complete, it converges to a point $p \in M$. By continuity of f, p is a fixed point of f. On the other hand, since f is a contraction, it has at most one fixed point. □

Exercise 2.1 (Perturbations of contractions)**.** Let $f\colon \mathbb{R}^d \circlearrowleft$ be a uniform contraction and $r > 0$ be arbitrary. We define the following balls

$$B_r^0(f) := \left\{ g \in C^0(\mathbb{R}^d, \mathbb{R}^d) : \|f - g\|_{C^0} := \sup_{x\in\mathbb{R}^d} \|f(x) - g(x)\| < r \right\},$$

and

$$B_r^L(f) := \left\{ g \in C^0(\mathbb{R}^d, \mathbb{R}^d) : \|f - g\|_L := \|f - g\|_{C^0} + \mathrm{Lip}(f - g) < r \right\}.$$

Show that given any $r > 0$, there is a $g \in B_r^0(f)$ which is not a contraction and such that $\mathrm{Fix}(g) := \{x \in \mathbb{R}^d : g(x) = x\}$ is an infinite set.

On the other hand, show that given the uniform contraction f, there exists $r_0 > 0$ such that every $g \in B_{r_0}^L(f)$ is a uniform contraction, and hence it exhibits a unique fixed point $p_g \in \mathbb{R}^d$. Finally, prove that given any $\varepsilon > 0$, there exists $\delta \in (0, r_0)$ such that

$$\|p_f - p_g\| < \varepsilon, \quad \forall g \in B_\delta^L(f).$$

Another important example is the following one:

Example 2.2 (Gradient flows)**.** Let M be the unit sphere $\mathbb{S}^2 := \{v \in \mathbb{R}^3 : \|v\| = 1\}$ and consider the function $L\colon \mathbb{S}^2 \to \mathbb{R}$ given by

$$L(x, y, z) := -z, \quad \forall (x, y, z) \in \mathbb{S}^2,$$

and the flow $\Phi\colon \mathbb{R} \times \mathbb{S}^2 \to \mathbb{S}^2$ generated by the vector field $X(v) := -\nabla L(v)$, for every $v \in \mathbb{S}^2$ and where ∇ denotes the gradient operator on $\mathbb{S}^2$.

One can easily show that $(0, 0, -1)$ and $(0, 0, 1)$ are the only singularities of X (i.e. the critical points of L), or equivalently, the fixed points of Φ, and it holds

$$\Phi^t(v) \to (0, 0, -1), \quad \text{and} \quad \Phi^{-t}(v) \to (0, 0, 1),$$

as $t \to +\infty$, for every $v \in \mathbb{S}^2 \backslash \{(0, 0, \pm 1)\}$.

Generalizing this example, one can consider an arbitrary compact differentiable manifold M endowed with a Riemannian structure $\langle \cdot, \cdot \rangle$ and a smooth function $L\colon M \to \mathbb{R}$. Then we can define the gradient of L as the only vector field ∇L such that

$$YL = dL(Y) = \langle \nabla L, Y \rangle,$$

for every vector field Y on M.

So, if we consider the flow $\Phi\colon \mathbb{R} \times M \to M$ induced by the vector field $X := -\nabla L$, one can easily check that

$$\partial_t (L \circ \Phi^t)(p)\big|_{t=0} = -\|\nabla L(p)\|^2 = -\langle \nabla L(p), \nabla L(p) \rangle, \quad \forall p \in M.$$

That means, given any point $p \in M$ the map $\mathbb{R} \ni t \mapsto L \circ \Phi^t(p) \in \mathbb{R}$ is not increasing, and it is strictly decreasing if and only if p is a regular point for L is a regular point of L, i.e. $dL_p \colon T_pM \to \mathbb{R}$ is not identically zero.

Moreover, if L has a finite number of critical points (i.e. points where the derivative dL vanishes), then one can show that for every $p \in M$, there exist critical points of L $\alpha_p, \omega_p \in M$ such that

$$\Phi^{-t}(p) \to \alpha_p, \qquad \Phi^t(p) \to \omega_p,$$

as $t \to +\infty$.

3. The Fundamental Theorem of Dynamical Systems

Motivated by Example 2.2, we would like to find functions decreasing along the orbits of arbitrary topological dynamical systems.

Definition 3.1 (Lyapunov function)**.** Given a continuous dynamical systems $f \colon M \circlearrowleft$ on a metric space M, a *Lyapunov function* for f is a continuous function $L \colon M \to \mathbb{R}$ such that

$$L\big(f(x)\big) \leqslant L(x), \quad \forall x \in M.$$

Of course, according to this definition every constant function is a Lyapunov one for every topological dynamical system, which does not seem to be very interesting. So, we are going to look for the "best possible" Lyapunov function.

3.1. **Different forms of recurrence.** It would be desirable to find a Lyapunov function such that $L\big(f(x)\big) < L(x)$ for every $x \in M$. However this is clearly impossible when f has, for instance, a **fixed point,** i.e. a point $p \in M$ such that $f(p) = p$. The set of fixed points of f will be denoted by $\mathrm{Fix}(f)$.

More generally, a point $p \in \mathrm{Per}(f)$ is said to be **periodic** when there is $n \geqslant 1$ such that $f^n(p) = p$. We define $\mathrm{Per}(f) := \bigcup_{n \geqslant 1} \mathrm{Fix}(f^n)$.

Another form of recurrence that represents an obstruction for the existence of a Lyapunov functions satisfying the strict inequality for every $p \in M$ is the following: we say that a point $p \in M$ is **recurrent** for f when there exists strictly increasing sequence of natural numbers $n_j \uparrow +\infty$ such that $f^{n_j}(p) \to p$, as $n_j \to +\infty$. The set of recurrent points of f shall be denoted by $\mathrm{Rec}(f)$.

Given a point $p \in M$, we define its **ω-limit set** by

$$\omega_f(p) := \{x \in M : \exists n_j \uparrow \infty,\ f^{n_j}(p) \to x, \text{ as } n_j \to +\infty\}.$$

When f is invertible, we also define the **α-limit set** of $p \in M$ by $\alpha_f(p) := \omega_{f^{-1}}(p)$, and the so called **limit set** of f by

$$L(f) := \bigcup_{p \in M} \alpha_f(p) \cup \omega_f(p).$$

We say that a point $p \in M$ is **non-wandering** when for every neighborhood U of p, there exists $n \geqslant 1$ such that $U \cap f^{-n}(U) \neq \varnothing$. The set of non-wandering points for f is usually denoted by $\Omega(f)$.

When $f \colon M \circlearrowleft$ is bijective, we say that $K \subset M$ is an **f-invariant set** when $f(K) = K$.

Exercise 3.1. Let M denote a compact metric space and $f \colon M \circlearrowleft$ be a homeomorphism. Show that

$$\mathrm{Fix}(f) \subset \mathrm{Per}(f) \subset \mathrm{Rec}(f) \subset L(f) \subset \Omega(f),$$

and all these sets are f-invariant.

(1) Show that $\mathrm{Fix}(f)$ and $\Omega(f)$ are compact.

(2) Show that $\mathrm{Rec}(f) \cap \mathrm{Rec}(f^{-1}) \neq \varnothing$. (Hint: show that there exists a non-empty compact subset $K \subset M$ which is **minimal** for f, i.e. K is f-invariant and it does not contain any proper compact f-invariant subset).
(3) Show that if $L\colon M \to \mathbb{R}$ is a Lyapunov function for f and $p \in \Omega(f)$, then it holds $L\big(f(p)\big) = L(p)$.

Finally we introduce the notion of *chain recurrence.* Given a homeomorphism $f\colon M \circlearrowleft$ on the compact metric space (M, d) and an arbitrary $\varepsilon > 0$, a sequence $(x_n)_{n\in\mathbb{Z}}$ of points of M is said to be an **ε-chain** or **ε-pseudo-orbit** when

$$d\big(f(x_i), x_{i+1}\big) < \varepsilon, \quad \forall n \in \mathbb{Z}.$$

A point $p \in M$ is said to be *chain recurrent* when for every $\varepsilon > 0$ there exists a an ε-chain $(x_n)_{n\in\mathbb{Z}}$ and $k \geqslant 1$ such that $p = x_0 = x_k$. The set of chain recurrent points is denoted by $\mathcal{R}(f)$.

Exercise 3.2. Show that $\mathcal{R}(f)$ is a compact f-invariant set and $\Omega(f) = \Omega(f^{-1}) \subset \mathcal{R}(f) = \mathcal{R}(f^{-1})$. On the other hand, show that if $L\colon M \to \mathbb{R}$ is a Lyapunov function for f and $p \in \mathcal{R}(f)$, then $L\big(f(p)\big) = L(p)$.

Given any $p \in M$ and any $\varepsilon > 0$, we define the **ε-chainable set from** p as the set

$$\mathcal{R}_f(p, \varepsilon) := \{x \in M : \exists (x_n)_{n\in\mathbb{Z}}\ \varepsilon\text{-chain}, \exists k \in \mathbb{N}, x_0 = p,\ x_k = x\},$$

and the **chainable set from** p by

$$\mathcal{R}_f(p) := \bigcap_{\varepsilon>0} \mathcal{R}_f(p, \varepsilon).$$

So, we have $p \in \mathcal{R}(f)$ if and only if $p \in \mathcal{R}_f(p)$. Then we have the following

Exercise 3.3 (Chain-recurrence classes)**.** Let $f\colon M \circlearrowleft$ be a homeomorphism on the compact metric space (M, d). Given two points $x, y \in \mathcal{R}(f)$, we say that they are **chain equivalent** when $y \in \mathcal{R}_f(x)$ and $x \in \mathcal{R}_f(y)$. Show this determines an equivalence relation on $\mathcal{R}(f)$. The equivalence classes of this relation are called **chain-recurrence classes** and prove that every chain-recurrent class is compact and f-invariant.

Now we can formulate the so called "Fundamental Theorem of Dynamical Systems" due to Conley [Con78]:

Theorem 3.1 (Existence of complete Lyapunov functions)**.** *Given a homeomorphism $f\colon M \circlearrowleft$ on the compact metric space (M, d), there exists a* complete Lyapunov function *for f, i.e. a continuous function $L\colon M \to \mathbb{R}$ such that the following conditions hold:*

(1) $L\big(f(x)\big) \leqslant L(x)$, for every $x \in M$,
(2) $L\big(f(x)\big) = L(x)$ if and only if $x \in \mathcal{R}(f)$,
(3) $L(x) = L(y)$, with $x, y \in \mathcal{R}(f)$ if and only if $\mathcal{R}_f(x) = \mathcal{R}_f(y)$,
(4) $L\big(\mathcal{R}(f)\big)$ is a compact totally disconnected set, i.e. it does not contain any open interval.

Now we are going to discuss some fundamental ideas and concepts that are needed for the proof of Theorem 3.1. See for instance [FM02] for a complete proof of this result.

A non-empty open set $U \subset M$ is said to be a **trapping region** for f when it hods $f(\overline{U}) \subset U$. Observe that the whole space M is, by definition, a trivial trapping region for f.

Lemma 3.2. *The ε-chainable set $\mathcal{R}_f(p, \varepsilon)$ is a trapping region for every $p \in M$ and every $\varepsilon > 0$.*

Proof. Let $\varepsilon > 0$ be an arbitrary number and $p \in M$ an arbitrary point. By its definition, it is clear the ε-chain set $\mathcal{R}_f(p,\varepsilon)$ is open and non-empty. On the other hand, since M is compact, f is uniformly continuous. So there is $\delta > 0$ such that $d\big(f(x), f(y)\big) < \varepsilon$, for every $x, y \in M$ with $d(x,y) < \delta$. We can assume $\delta < \varepsilon$.

Now, let $y \in \overline{\mathcal{R}_f(p,\varepsilon)}$ be an arbitrary point. We want to show that $f(y) \in \mathcal{R}_f(p,\varepsilon)$. Since $y \in \overline{\mathcal{R}_f(p,\varepsilon)}$, there exists z in the open ball $B_\delta(y) \cap \mathcal{R}_f(p)$ and thus, an ε-chain $(x_n)_{n\in\mathbb{Z}}$ with $x_0 = p$ and $x_k = z$. Since $d(y,z) < \delta$, we get $d\big(f(y), f(z)\big) < \varepsilon$ and hence $x_0 = p, x_1, \dots, x_k = z, f(y)$ is a new ε-chain and hence, $f(y) \in \mathcal{R}_f(p)$. □

Then, given a trapping region $U \subset M$ we can define its associated **attracting set** $A_U \subset M$ by

$$A_U := \bigcap_{n\geqslant 0} f^n(U).$$

Analogously, when $\overline{U} \neq M$, the non-empty open set $M\backslash\overline{U}$ is a trapping region for f^{-1}. So we can define the **dual repelling set** $A_U^\star \subset M$ by

$$A_U^\star := \bigcap_{n\geqslant 0} f^{-n}\big(M\backslash\overline{U}\big).$$

Exercise 3.4. Show A_U and $A_U^\star$ are compact, non-empty and f-invariant (i.e. $f(A_U) = A_U$ and $f(A_U^\star) = A_U^\star$), and it holds $\mathcal{R}(f) \subset A_U \cup A_U^\star$.

The pair $(A_U, A_U^\star)$ is called the **attracting-reppeling pair** associated to U.

Then, we finally need the following lemmas:

Lemma 3.3. *The set $\{(A_U, A_U^\star)$ attracting-repelling pair$\}$ is either finite or countable.*

Lemma 3.4. *We have*

$$\mathcal{R}(f) = \bigcap_{(A_U, A_U^\star)} (A_U \cup A_U^\star),$$

where the intersection is taken over all attracting-repelling pairs.

Lemma 3.5. *Given trapping region $U \subset M$ with $\overline{U} \neq M$, there exists a Lyapunov function $L\colon M \to [0,1]$ for f such that $f\big|_{A_U} \equiv 0$, $f\big|_{A_U^\star} \equiv 1$ and it hods*

$$0 < L(f(x)) < L(x) < 1, \quad \forall p \in M\backslash(A_U \cup A_U^\star).$$

Proof of Lemma 3.3. It is easily follows from the fact that, since M is compact, its topology admits a countable base. □

Proof of Lemma 3.4. It follows from Lemma 3.2 and Exercise 3.4. □

Proof of Lemma 3.5. It is left as an exercise. □

Exercise 3.5. If M is a compact metric space and $f\colon M \circlearrowleft$ is a homeomorphism, show that

$$\mathcal{R}\left(f\big|_\Lambda\right) = \mathcal{R}(f) \cap \Lambda,$$

for every compact f-invariant set $\Lambda \subset M$, i.e. where $f(\Lambda) = \Lambda$.

On the other hand, give an example where $\Omega\big(f\big|_{\Omega(f)}\big) \neq \Omega(f)$.

4. Dynamics of homeomorphisms on one-dimensional manifolds

In this section we start the study of the dynamics of homeomorphisms and diffeomorphisms of compact manifolds. To do that, we start considering one-dimensional manifolds, recalling that up to diffeomorphisms, there are only to compact connected one-dimensional manifolds: the closed interval and the circle (see Milnor [Mil65] for a proof of this fact).

4.1. **Homeomorphisms of the interval.** Let $I \subset \mathbb{R}$ be a compact non-trivial interval, i.e. a compact connected subset of the real line that does not reduce to a point.

Given any homeormophism $f\colon I \circlearrowleft$, it can either preserve or revert orientation, or equivalently, it can be either stricly increasing or stricly decreasing, respectively.

4.1.1. *Orientation preserving interval homeomorphisms.* If $f\colon I \circlearrowleft$ is a orientation preserving homeormophism, one can easy show that $\partial I \subset \mathrm{Fix}(f)$. In particular, $\mathrm{Fix}(f)$ is not empty and, moreover, it holds $\mathcal{R}(f) = \mathrm{Fix}(f)$.

In order to describe all "different dynamics" that can be found within $\mathrm{Homeo}_+(I)$, the group of orientation preserving homeomorphisms of the interval I, we first need define the notion of isomorphism in the category of discrete time dynamical systems:

Definition 4.1 (Topological and smooth conjugacy)**.** If M and N are compact metric spaces, we say that two homeomorphisms $f\colon M \circlearrowleft$ and $g\colon N \circlearrowleft$ are **topologically conjugate** when there exists a homeormophism $h\colon M \to N$ such that $h \circ f = g \circ h$.

When M and N are differentiable manifolds and f and g are C^r-diffeomorphisms, then we say f and g are **C^s-conjugate,** with $0 \leqslant s \leqslant r$ when there is a C^s-diffeomorphisms $h\colon M \to N$ satisfying $h \circ f = g \circ h$.

Exercise 4.1. Let $f\colon M \circlearrowleft$ and $g\colon N \circlearrowleft$ be homeomorphisms and $h\colon M \to N$ be homeomorphism that conjugates them. Show that $h\big(\mathrm{Fix}(f)\big) = \mathrm{Fix}(g)$, $h\big(\mathrm{Per}(f)\big) = \mathrm{Per}(g)$, $h\big(\mathrm{Rec}(f)\big) = \mathrm{Rec}(g)$, $h\big(L(f)\big) = L(g)$, $h\big(\Omega(f)\big) = \Omega(g)$ and $h\big(\mathcal{R}(f)\big) = \mathcal{R}(g)$.

Exercise 4.2. Let $f, g \in \mathrm{Homeo}_+(I)$ be orientation perserving homeomorphisms such that $\mathrm{Fix}(f) = \mathrm{Fix}(g) = \partial I$. Show that there are topologically conjugate.

Give an example of $f, g \in \mathrm{Diff}^\infty_+(I)$, i.e. orientation preserving C^∞-diffeomorphisms such that $\mathrm{Fix}(f) = \mathrm{Fix}(g) = \partial I$, but they are not C^1-conjugate.

Finally, give an example of diffeomorphisms $f, g \in \mathrm{Diff}^\infty_+(I)$ with $\mathrm{Fix}(f) = \mathrm{Fix}(g)$, which are not toplogically conjugate.

The notion of **structural stability** is another fundamental one in Dynamical Systems, that is strongly related to the phenomenon of *hyperbolicity.* We say that a C^r-diffeomorphism $f \in \mathrm{Diff}^r(I)$ is **C^s-structural stable,** with $0 \leqslant s \leqslant r$, when there is an $\varepsilon > 0$ such that g is *topologically conjugate* to f, for every $g \in B^s_\varepsilon(f)$, where

$$\mathcal{V}^s_\varepsilon(f) := \left\{ g \in \mathrm{Diff}^s(I) : \max_{x \in I} \max_{0 \leqslant i \leqslant s} \left| D^i f(x) - D^i g(x) \right| < \varepsilon \right\}.$$

Definition 4.2 (Hyperbolic fixed points)**.** On the other hand, given a C^1-diffeomorphism $f\colon I \circlearrowleft$, a fixed point $p \in I$ of f is said to be **hyperbolic** when $|Df(p)| \neq 1$. A hyperbolic fixed point p is said to be **attracting** when $|Df(p)| < 1$ and **repelling** when $|Df(p)| > 1$.

Exercise 4.3. show that hyperbolic fixed points are isolated points of the fixed point set. In other words, show that given a C^1-diffeomorphism $f\colon I \circlearrowleft$ and a hyperbolic fixed point $p \in I$, there exists $r > 0$ such that

$$B_r(p) \cap \mathrm{Fix}(f) = \{p\},$$

where $B_r(p)$ denotes the open ball $\{x \in I : |x - p| < r\}$.

If p is an attracting hyperbolic fixed point, show that there exists $\varepsilon > 0$ such that $\omega_f(x) = \{p\}$, for every $x \in B_\varepsilon(p)$. Conversely, if p is a repelling hyperbolic point, show that there exists $\varepsilon > 0$ satisfying the following properties:

- $\alpha_f(x) = \{p\}$, for every $x \in B_\varepsilon(p)$;
- for every $y \in B_\varepsilon(p) \backslash \{p\}$, there exists $n > 0$ such that $f^n(y) \notin B_\varepsilon(p)$.

Theorem 4.1 (Structural stability for interval diffeomorphisms)**.** *A diffeomorphisms $f \in \mathrm{Diff}^1_+(I)$ is C^1-structural stable if and only if every fixed point of f is hyperbolic.*

On the other hand, for every $r \in \mathbb{N}$ the set of structural stable diffeomorphisms

$$\mathcal{H}^r_+(I) := \left\{ f \in \mathrm{Diff}^r_+(I) : \forall p \in \mathrm{Fix}(f),\ p \text{ is hyperbolic} \right\}$$

is an open and dense subset of $\mathrm{Diff}^r_+(I)$, which is endowed with the uniform C^r-topology.

4.1.2. *Orientation-reversing homeomorphisms.* Given an orientation-reversing homeomorphism $f\colon I \circlearrowleft$ one can easily show that it has exactly one fixed-point, the boundary points of I are the elements of a periodic orbit of **prime period** 2, i.e. $f(x) \neq x = f^2(x)$ for each $x \in \partial I$, and any other periodic point has prime period 2, too. One can also show that $\mathcal{R}(f) = \mathrm{Per}(f)$.

On the other hand, we can naturally extend the notion of hyperbolicity for periodic points of C^1-diffeomorphisms, saying that a periodic point p is **hyperbolic** when $\left|Df^k(p)\right| \neq 1$ and $f^k(p) = p$, with $k \geqslant 1$.

Finally, one can also show that a C^1 orientation reversing diffeomorphisms $f\colon I \circlearrowleft$ is C^1-structural stable if and only if all its periodic orbits are hyperbolic and the set

$$\mathcal{H}^r_-(I) := \left\{ f \in \mathrm{Diff}^r_-(I) : \forall p \in \mathrm{Per}(f), p \text{ is hyperbolic} \right\}$$

is an open and dense subset of $\mathrm{Diff}^r_-(I)$.

4.2. **Circle homeomorphisms.** In general, there are four usual models for the circle: it can be either defined as the sub-manifold of $\mathbb{R}^2$ given by $\mathbb{S}^1 := \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$; or as a subset of $\mathbb{C}$ given by $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$; or as the real projective line $\mathbb{R}\mathrm{P}^1 := \mathbb{R}^2\backslash\{0\}/\sim$, where $v \sim w$ for $v, w \in \mathbb{R}^2\backslash\{0\}$ if and only if $\exists \lambda \in \mathbb{R}$ with $\lambda v = w$; or as the quotient space $\mathbb{T} := \mathbb{R}/\mathbb{Z}$. In all these cases they can be endowed with natural topologies, metrics, differentiable structures and group structures, and all them are equivalent, as they are related, for instance, by the following maps: $D\colon \mathbb{T} \to \mathbb{S}^1 \subset \mathbb{C}$ given by

$$D(t + \mathbb{Z}) := e^{2\pi i t}, \quad \forall t \in \mathbb{R}.$$

Let us introduce a notation that will be very useful along these notes. Given a topological space X and any point $x_\star \in X$, the pair $(X, x_\star)$ is called a **pointed space.** Then, a **pointed map** of pointed spaces is an object $f\colon (X, x_\star) \to (Y, y_\star)$, where $f\colon X \to Y$ is a continuous map satisfying $f(x_\star) = y_\star$.

Returning to our discussion of circle maps, here we shall use the model $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ for the circle, but this is just an arbitrary choice. The natural quotient map $\pi\colon \mathbb{R} \to \mathbb{T}$ given by $\pi(t) := t = t + \mathbb{Z}$, for every $t \in \mathbb{R}$ turns to be the **universal cover** of the circle. So, given any continuous map $f\colon \mathbb{T} \circlearrowleft$ there exists a **lift** of f (which is never unique) $\tilde{f}\colon \mathbb{R} \circlearrowleft$ that is continuous and satisfies $\pi \circ \tilde{f} = f \circ \pi$.

More precisely, one can show that given a (continuous) pointed map $f\colon (\mathbb{T}, x_\star) \to (\mathbb{T}, y_\star)$ and arbitrary points $\tilde{x}_\star \in \pi^{-1}(x_\star) \subset \mathbb{R}$ and $\tilde{y}_\star \in \pi^{-1}(y_\star) \subset \mathbb{R}$, there exists a unique lift of f which is a pointed map $\tilde{f}\colon (\mathbb{R}, \tilde{x}_\star) \to (\mathbb{R}, \tilde{y}_\star)$.

A continuous map $f\colon \mathbb{T} \circlearrowleft$ is said to be a **local homeomorphism** when for every $y \in \mathbb{T}$ and every $x \in f^{-1}(y)$ there exist neighborhoods U and V of x and y, respectively, such that $f\big|_U\colon U \to V$ is a homeomorphism.

Exercise 4.4. Let $f\colon \mathbb{T} \circlearrowleft$ be a continuous map and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f. Show that f is a local homeomorphism if and only if $\tilde{f}$ is a homeomorphism.

One can easily show that given two lifts $\tilde{f}, \tilde{g}\colon \mathbb{R} \circlearrowleft$ of the continuous map $f\colon \mathbb{T} \circlearrowleft$, there exists an integer number k such that $\tilde{f}(\tilde{x}) = \tilde{g}(\tilde{x}) + k$, for every $\tilde{x} \in \mathbb{R}$.

The **topological degree** of a continuous map $f\colon \mathbb{T} \circlearrowleft$ is the integer number $\deg(\tilde{f}) \in \mathbb{Z}$ given by

$$\deg(f) := \tilde{f}(\tilde{x}+1) - \tilde{f}(\tilde{x}), \tag{4}$$

for every $\tilde{x} \in \mathbb{R}$ and any lift $\tilde{f}\colon \mathbb{R} \circlearrowleft$ of f.

Exercise 4.5. Show that the real number $\deg(f)$ given by (4) is indeed an integer number, which does neither depend on the point $\tilde{x}$ nor on the lift $\tilde{f}$.

Exercise 4.6. Given continuous functions $f, g\colon \mathbb{T} \circlearrowleft$, show that

$$\deg(f \circ g) = \deg(f)\deg(g).$$

Use this equality to prove that if $f\colon \mathbb{T} \circlearrowleft$ is a homeormophism, then it holds $\deg(f) \in \{-1, 1\}$.

Finally, if $f\colon \mathbb{T} \circlearrowleft$ is an arbitrary continuous function and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift of f, show that the continuous function $\Delta_{\tilde{f}}\colon \mathbb{R} \to \mathbb{R}$ given by

$$\Delta_{\tilde{f}}(\tilde{x}) := \tilde{f}(\tilde{x}) - \deg(f)\tilde{x}, \quad \forall \tilde{x} \in \mathbb{R},$$

is $\mathbb{Z}$-periodic and hence, we can consider $\Delta_{\tilde{f}} \in C^0(\mathbb{T}, \mathbb{R})$.

Similarly to interval homeomorphisms, the family of circle homeomorphisms is divided in orientation preserving and orientation reversing ones: given a homeomorphism $f\colon \mathbb{T} \circlearrowleft$ we say that f is **orientation preserving** when $\deg(f) = 1$ and is **orientation reversing** when $\deg(f) = -1$.

4.2.1. *Orientation-reversing homeomorphisms.* Let $f\colon \mathbb{T} \circlearrowleft$ be an orientation-reversing homeomorphism. Then f has exactly two fixed points and any other periodic orbit has prime period equal to 2.

Exercise 4.7. Show that if $f\colon \mathbb{T} \circlearrowleft$ is an orientation-reversing homeomorphism, then it holds $\sharp\,\mathrm{Fix}(f) = 2$ and

$$\mathrm{Per}(f) = \mathrm{Fix}(f^2) = \Omega(f).$$

Give an example where $\mathrm{Per}(f) \neq \mathcal{R}(f)$.

4.2.2. *Orientation-preserving circle homeomorphisms.* Let us start considering a important family of orientation preserving circle diffeomorphisms which are usually called **Möbious transformations.** To do that, we shall use the projective line model $\mathbb{R}\mathrm{P}^1$ to represent the circle. Recall that the d-dimensional projective space $\mathbb{R}\mathrm{P}^d$ is defined as $\mathbb{R}^{d+1}\backslash\{0\}/\sim$, where $v \sim w$ with $v, w \in \mathbb{R}^d\backslash\{0\}$ if and only if there is $\lambda \in \mathbb{R}$ with $\lambda v = w$. The one-dimensional projective is diffeomorphic to $\mathbb{T}$ by the map $D_1\colon \mathbb{T} \to \mathbb{R}\mathrm{P}^1$ given by

$$D_1(t + \mathbb{Z}) := \Big[\big(\cos(\pi t), \sin(\pi t)\big)\Big], \quad \forall t \in \mathbb{R},$$

and where $[v]$ denotes the equivalence class of the vector $v \in \mathbb{R}^2\backslash\{0\}$. On the other hand, let us write $\mathfrak{gl}_d(\mathbb{R})$ for the space of $d \times d$ matrices with real coefficients and recall that each $A \in \mathfrak{gl}_d(\mathbb{R})$ naturally induces a projective map $[A]\colon \mathbb{R}\mathrm{P}^d \circlearrowleft$ by $[A][v] := [Av]$, for every $v \in \mathbb{R}^d\backslash\{0\}$. This map $[A]$ is a diffeomorphism if and only if $\det A \neq 0$. So, if we define the d-dimensional special linear group $\mathrm{SL}_d(\mathbb{R}) := \{A \in \mathfrak{gl}_d(\mathbb{R}) : \det A = 1\}$, a **Möbious transformation** is an orientation preserving circle diffeomorphism $[A]\colon \mathbb{R}\mathrm{P}^1 \circlearrowleft$, with $A \in \mathrm{SL}_2(\mathbb{R})$.

The dynamics of Möbious transformations is completely characterized by the spectral properties of the corresponding linear transformation. In fact, given any $A \in \mathrm{SL}_2(\mathbb{R})$, let us write $\lambda_1, \lambda_2 \in \mathbb{C}$ for its corresponding complex eigenvalues. So, the matrix A can either be the identity (i.e. $A = I$), or can be **elliptic** (where

$\lambda_1 = \bar{\lambda}_2 \in \mathbb{C}\backslash\mathbb{R}$), or **parabolic** (where $\lambda_1 = \lambda_2 = \pm 1$), or **hyperbolic** (where $|\lambda_1| > 1$ and $\lambda_1 = \lambda_2^{-1} \in \mathbb{R}$).

When A is elliptic, by Jordan normal form theorem we know that there exist $B \in \mathrm{SL}_2(\mathbb{R})$ and $\alpha \in \mathbb{R}$ such that $A = BR_\alpha B^{-1}$, where

$$R_\alpha := \begin{pmatrix} \cos(2\pi\alpha) & -\sin(2\pi\alpha) \\ \sin(2\pi\alpha) & \cos(2\pi\alpha) \end{pmatrix}.$$

That means that A is linearly conjugate to the **rotation** R_α, and since $[A] = [B] \circ [R_\alpha] \circ [B]^{-1}$, we conclude that the corresponding Möbious transformations $[A]$ and $[R_\alpha]$ are indeed conjugate.

Exercise 4.8. Given $\alpha \in \mathbb{Q}$, show that $\mathrm{Per}([R_\alpha]) = \mathrm{Fix}([R_\alpha]^q) = \mathbb{R}\mathrm{P}^1$, where $\alpha = p/q$, with $q \in \mathbb{N}$, $p \in \mathbb{Z}$ and $\gcd(p, q) = 1$.

On the other hand, show that if $\alpha\mathbb{R}\backslash\mathbb{Q}$, then $[R_\alpha]$ is **minimal,** i.e. every orbit is dense in $\mathbb{R}\mathrm{P}^1$.

When A is parabolic, one can show that $[A]$ has exactly one fixed point $[v] \in \mathbb{R}\mathrm{P}^1$, having

$$\{[v]\} = \mathrm{Fix}([A]) = \mathrm{Per}([A]) = \mathrm{Rec}([A]) = L([A]) = \Omega([A]) \neq \mathcal{R}([A]) = \mathbb{R}\mathrm{P}^1.$$

Exercise 4.9. Show that, $[A]'([v]) = 1$.

Finally, when A is hyperbolic the corresponding Möbious map $[A]\colon \mathbb{R}\mathrm{P}^1 \circlearrowleft$ has exactly two fixed points $[v_s], [v_u] \in \mathbb{R}\mathrm{P}^1$, with $[A]'[v_s] = |\lambda_1|$ and $[A]'[v_u] = |\lambda_2|$. In this case it holds $\mathrm{Fix}([A]) = \mathcal{R}([A])$.

So, now let us start the study of the dynamical properties of arbitrary orientation preserving circle homeomorphisms. Let $f\colon \mathbb{T} \circlearrowleft$ be a homeomorphisms with $\deg(f) = 1$ and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f. Then we have that $\Delta_{\tilde{f}} := \tilde{f} - id_{\mathbb{R}}$ is $\mathbb{Z}$-periodic and hence can be considered as a real continuous function defined on $\mathbb{T}$.

Exercise 4.10. Show that if $f\colon \mathbb{T} \circlearrowleft$ is an orientation preserving homeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift, then it holds

$$\Delta_{\tilde{f}^n} = \sum_{j=0}^{n-1} \Delta_{\tilde{f}} \circ f^j, \quad \forall n \in \mathbb{N}.$$

Can you get an analogous formula for $n < 0$?

Now we introduce the most important dynamical invariant associated to (orientation preserving) circle homeomorphisms, the so called **rotation number,** whose definition is originally due to Poincaré:

Definition 4.3 (Rotation number)**.** Given an orientation preserving homeomorphism $f\colon \mathbb{T} \circlearrowleft$ and a lift $\tilde{f}\colon \mathbb{R} \circlearrowleft$, the *rotation number* is defined by

$$\rho(\tilde{f}) := \lim_{n \to +\infty} \frac{\tilde{f}^n(\tilde{x}) - x}{n},$$

for any $\tilde{x} \in \mathbb{R}$.

We immediately need the following

Theorem 4.2. *The limit defining the rotation number of $\tilde{f}$ exists and is independent of $\tilde{x}$, and the convergence is in deed uniform on $\tilde{x}$.*

The following simple lemma will play a crucial role in the proof of Theorem 4.2 and subsequent results:

Lemma 4.3 (Oscillation of $\Delta_{\tilde{f}}$)**.** *If $f\colon \mathbb{T} \circlearrowleft$ is an orientation preserving homeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift of f, then it holds*

$$\operatorname{osc} \Delta_{\tilde{f}} := \max_{x\in\mathbb{T}} \Delta_{\tilde{f}}(x) - \min_{x\in\mathbb{T}} \Delta_{\tilde{f}}(x) < 1.$$

Proof of Lemma 4.3. Since $\Delta_{\tilde{f}}$ is $\mathbb{Z}$-periodic when considered as a real function defined on $\mathbb{R}$, then there exists numbers $x_m, x_M \in \mathbb{R}$ such that $x_m < x_M < x_m + 1$ and $\Delta_{\tilde{f}}(x_m) = \min \Delta_{\tilde{f}}$ and $\Delta_{\tilde{f}}(x_M) = \max \Delta_{\tilde{f}}$. Reasoning by contradiction, let us suppose that $\operatorname{osc} \Delta_{\tilde{f}} \geqslant 1$. Then it holds

$$\tilde{f}(x_M) - \tilde{f}(x_m) = x_M - x_m + \Delta_{\tilde{f}}(x_M) - \Delta_{\tilde{f}}(x_m) > 1.$$

Thus, there exists $y \in \mathbb{R}$ such that $\tilde{f}(x_m) < y < y + 1 < \tilde{f}(x_M)$. By continuity of $\tilde{f}$, there exists two points $x_0, x_1 \in \mathbb{R}$, such that $x_m < x_0 < x_1 < x_M$, and $\tilde{f}(x_0) = y$ and $\tilde{f}(x_1) = y + 1$. Then, we have

$$f\big(\pi(x_i)\big) = \pi\big(\tilde{f}(x_i)\big) = \pi(y + i) = \pi(y),$$

for $i = 0, 1$, but since $|x_1 - x_0| < x_M - x_m < 1$, it holds $\pi(x_0) \neq \pi(x_1)$, contradicting the injectivity of f. □

We also need the following elementary facts

Lemma 4.4. *Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers such that there is a real constant $C > 0$ satisfying*

$$|a_{m+n} - a_m - a_n| \leqslant C, \quad \forall m, n \in \mathbb{N}.$$

Then, the following limit exists

$$\rho := \lim_{n\to+\infty} \frac{a_n}{n},$$

and it holds $|a_n - n\rho| \leqslant C$, for every $n \in \mathbb{N}$.

Lemma 4.5. *Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers such that*

$$a_{m+n} \leqslant a_m + a_n, \quad \forall m, n \in \mathbb{N}.$$

Then, the there exists $-\infty \leqslant \rho < +\infty$ such that

$$\rho = \lim_{n\to+\infty} \frac{a_n}{n} = \inf_{n\geqslant 1} \frac{a_n}{n}$$

Proofs of Lemmas 4.4 and 4.5. Both are left as exercises. □

Proof of Theorem 4.2. As a consequence of Exercise 4.10 Lemma 4.3 we easily get

$$\begin{aligned}
&\big|\tilde{f}^{m+n}(0) - \tilde{f}^m(0) - \tilde{f}^n(0)\big| \\
&= \left|\sum_{i=0}^{m+n-1} \Delta_{\tilde{f}}\big(\tilde{f}^i(0)\big) - \sum_{i=0}^{m-1} \Delta_{\tilde{f}}\big(\tilde{f}^i(0)\big) - \sum_{i=0}^{n-1} \Delta_{\tilde{f}}\big(\tilde{f}^i(0)\big)\right| \\
&= \left|\sum_{i=0}^{n-1} \Delta_{\tilde{f}}\Big(\tilde{f}^i\big(\tilde{f}^m(0)\big)\Big) - \sum_{i=0}^{n-1} \Delta_{\tilde{f}}\big(\tilde{f}^i(0)\big)\right| = \left|\Delta_{\tilde{f}^n}\big(\tilde{f}^m(0)\big) - \Delta_{\tilde{f}^n}(0)\right| \\
&\leqslant \operatorname{osc} \Delta_{\tilde{f}^n} < 1.
\end{aligned}$$

So, by Lemma 4.4 we know that the limit

$$\lim_{n\to+\infty} \frac{\tilde{f}^n(0)}{n}$$

exists. Finally, in order to see that the limit defining the rotation number of $\tilde{f}$ is independent of $\tilde{x}$, and in fact, it is uniform on $\tilde{x}$, it is enough to verify that

$$\left|\frac{\tilde{f}^n(x) - x}{n} - \frac{\tilde{f}^n(0)}{n}\right| = \left|\frac{\Delta_{\tilde{f}^n}(x) - \Delta_{\tilde{f}^n}(0)}{n}\right| \leqslant \frac{\operatorname{osc} \Delta_{\tilde{f}^n}}{n} \leqslant \frac{1}{n},$$

where the last inequality is consequence of Lemma 4.3. □

Some fundamental properties of the rotation number are given by the following

Proposition 4.6. *If $f\colon \circlearrowleft$ is an orientation preserving homeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ is a lift of f, then the following properties hold:*

(1) $\rho(\tilde{f}^q - p) = q\rho(\tilde{f}) - p$, for every $p, q \in \mathbb{Z}$;
(2) $\forall n \in \mathbb{Z}$, $\exists x_n \in \mathbb{T}$ such that $\Delta_{\tilde{f}^n}(x_n) = n\rho(\tilde{f})$;
(3) $\left|\tilde{f}^n(\tilde{x}) - \tilde{x} - n\rho(\tilde{f})\right| \leqslant 1$, for all $\tilde{x} \in \mathbb{R}$ and every $n \in \mathbb{Z}$;
(4) the map $\rho_r\colon \mathbb{R} \to \mathbb{R}$ given by $\rho_r : \mathbb{R} \ni t \mapsto \rho(\tilde{f} + t) \in \mathbb{R}$ is continuous.

Proof. The proof of (1) is left as an exercise. To prove (2), reasoning by contradiction let us suppose there is $n \in \mathbb{Z}$ such that $\Delta_{\tilde{f}^n}(x) \neq n\rho(\tilde{f})$, for every $x \in \mathbb{T}$. Without loss of generality, we can assume there is $\varepsilon > 0$ such that $\Delta_{\tilde{f}^n}(x) > n\rho(\tilde{f}) + \varepsilon$, for every $x \in \mathbb{T}$. This implies that

$$\rho(\tilde{f}^n) = \lim_{k\to+\infty} \frac{\tilde{f}^{nk}(\tilde{x}) - \tilde{x}}{k} = \lim_{k\to+\infty} \frac{1}{k} \sum_{i=0}^{k-1} \Delta_{\tilde{f}^n}\big(f^{in}(x)\big) \geqslant n\rho(\tilde{f}) + \varepsilon,$$

for every $\tilde{x} \in \mathbb{R}$ and $x = \pi(\tilde{x})$, contradicting (1).

Then, estimate (3) is consequence of Lemma 4.3 and (2). In fact, Given any $\tilde{x} \in \mathbb{R}$ and any $n \in \mathbb{Z}$, by (2) we know that there is $x_n \in \mathbb{T}$ such that $\Delta_{\tilde{f}^n}(x_n) = n\rho(\tilde{f})$. Then, by Lemma 4.3 we know that

$$\left|\tilde{f}^n(\tilde{x}) - \tilde{x} - n\rho(\tilde{f})\right| = \left|\Delta_{\tilde{f}^n}(\tilde{x}) - \Delta_{\tilde{f}^n}(x_n)\right| \leqslant \operatorname{osc} \Delta_{\tilde{f}^n} < 1.$$

Finally, let us see that (4) follows from (3). First observe that is enough to prove the continuity of ρ_r at $0 \in \mathbb{R}$. Then, invoking the natural order of $\mathbb{R}$ and the fact that $\tilde{f}$ preserves this order, one can easily check that $\rho_r(t) \geqslant \rho_r(s)$, for every $t > s$. So, the right limit $\rho_r(0+) := \lim_{t\downarrow 0} \rho_r(t)$ exists and it clearly holds $\rho_r(0+) \geqslant \rho_r(0)$. Reasoning by contraction, let us suppose $\rho_r(0+) > \rho_r(0)$. Then there is $n \in \mathbb{N}$ such that $n\big(\rho_r(0+) - \rho_r(0)\big) > 3$. Fixing this n, there is $t > 0$ sufficiently small such that

$$\left|\big(T_t \circ \tilde{f}\big)^n(\tilde{x}) - \tilde{f}^n(\tilde{x})\right| \leqslant \frac{1}{2}, \quad \forall \tilde{x} \in \mathbb{R}.$$

Then we get a contradiction noticing that

$$\begin{aligned} 2 &\geqslant \left|\left|\big(T_t \circ \tilde{f}\big)^n(\tilde{x}) - \tilde{x} - n\rho_r(t)\right| - \left|\tilde{f}^n(\tilde{x}) - \tilde{x} - n\rho_r(0)\right|\right| \\ &\geqslant n\rho_r(t) - n\rho_r(0) - \left|\big(T_t \circ \tilde{f}\big)^n(\tilde{x}) - \tilde{f}^n(\tilde{x})\right| \geqslant 3 - \frac{1}{2} = 2.5. \end{aligned}$$

So, $\rho_r(0+)$ must be equal to $\rho_r(0)$. Analogously, one can prove the left continuity of ρ_r at 0.

□

In order to state the main result about the crucial role the rotation number plays in the description of the dynamics of circle homeomorphisms, we first need to introduce the following

Definition 4.4 (Semi-conjugacy)**.** Given $f\colon M \circlearrowleft$ and $g\colon N \circlearrowleft$ continuous maps on the compact metric spaces M and N, we say that they are **semi-conjugate** when there is a continuous surjective map $h\colon M \to N$ such that $h \circ f = g \circ h$. In such a case, we also say that f is an **extension** of g and that g is a **factor** of f.

A mathetical object associated to dynamical systems is said to be a dynamical invariant when it coincides on systems that are conjugate or semi-conjugate. The following exercise shows that the rotation number is indeed a dynamical invariant:

Exercise 4.11. Let $f, g \in \mathrm{Homeo}_+(\mathbb{T})$ be given and suppose there is a continuous map $h\colon \mathbb{T} \circlearrowleft$ with $\deg h = 1$ such that $f \circ h = h \circ g$. Then, show that there exist $\tilde{f}\colon \mathbb{R} \circlearrowleft$ and $\tilde{g}\colon \mathbb{R} \circlearrowleft$ lifts of f and g, respectively, such that $\rho(\tilde{f}) = \rho(\tilde{g})$.

Theorem 4.7. *Let $f\colon \mathbb{T} \circlearrowleft$ be an orientation preserving circle homeomorphism and $\tilde{f}\colon \mathbb{R} \circlearrowleft$ be a lift of f. Then, it holds:*

(1) $\mathrm{Per}(f) \neq \varnothing$ *if and only if* $\rho(\tilde{f}) \in \mathbb{Q}$*, and if* $\rho(\tilde{f}) = p/q$ *with* p, q *coprime, then* $\mathrm{Per}(f) = \mathrm{Fix}(f^q) = \Omega(f)$.

(2) If $\rho(\tilde{f}) \in \mathbb{R}\backslash\mathbb{Q}$*, then there exists a unique minimal set for f and f is semi-conjugate to the rotation* $T_{\rho(\tilde{f})} : \mathbb{T} \ni \tilde{x} + \mathbb{Z} \mapsto \tilde{x} + \rho(\tilde{f}) + \mathbb{Z}$*. If f is minimal, then f and* $T_{\rho(\tilde{f})}$ *are indeed topologically conjugate.*

Proof of Theorem 4.7. If $x_0 \in \mathrm{Per}(f)$ and $\tilde{x}_0 \in \pi^{-1}(x) \subset \mathbb{R}$, then there clearly are numbers $q \in \mathbb{N}$ and $p \in \mathbb{Z}$ such that $\tilde{f}^q(\tilde{x}_0) = \tilde{x}_0 + p$. So we have $\tilde{f}^{qn}(\tilde{x}_0) = \tilde{x}_0 + np$, for every $n \in \mathbb{Z}$ and hence,

$$\rho(\tilde{f}) = \lim_{n\to+\infty} \frac{\tilde{f}^{nq}(\tilde{x}_0) - x_0}{nq} = \frac{p}{q} \in \mathbb{Q}.$$

Reciprocally, if $\rho(\tilde{f}) = p/q$, with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ coprime, then (2) of Proposition 4.6 we know that there is $x_q \in \mathbb{T}$ such that $\Delta_{\tilde{f}^q}(x_q) = q\rho(\tilde{f}) = p \in \mathbb{Z}$. So, $x_q \in \mathrm{Fix}(f^q) \subset \mathrm{Per}(f)$. Since any periodic point of an orientation preserving interval homeomorphism is fixed and they are the only non-wandering points, we conclude that $\mathrm{Fix}(f^q) = \mathrm{Per}(f) = \Omega(f)$, whenever $\rho(\tilde{f}) = p/q$ with p and q coprime.

Now, let us suppose $\rho(\tilde{f}) \in \mathbb{R}\backslash\mathbb{Q}$ and let $K_f \subset \mathbb{T}$ be a minimal set for f. If $K_f = \mathbb{T}$, then it is clearly the only minimal set for f. So, let us suppose there exists another minimal set $K \subset \mathbb{T}$ for f which is different from K_f. By minimality, it holds $K \cap K_f = \varnothing$ and $d(K_f, K) := \sup\{d(x,y) : x \in K_f,\ y \in K\} = \delta > 0$. Let I be a connected component of $\mathbb{T}\backslash K_f$ such that $I \cap K \neq \varnothing$. Since f has no periodic point, we know that I is a wandering set for f, i.e. $f^m(I) \cap f^n(I) = \varnothing$, for every $m, n \in \mathbb{Z}$ and $m \neq n$. Hence, $\sum_{m\in\mathbb{Z}} \mathrm{Leb}\big(f^m(I)\big) \leqslant 1$, and consequently, we have $\mathrm{Leb}\big(f^m(I)\big) \to 0$, as $m \to \infty$. So, there is $m \in \mathbb{N}$ such that $\mathrm{Leb}\big(f^m(I)\big) \leqslant \delta/2$ and since $\varnothing \neq f^m(K \cap I) = K \cap f^m(I)$, we get a contradiction with the fact that $d(K_f, K) = \delta$, noticing that the extreme point of $f^m(I)$ belong to K_f.

Finally, in order to show that f is a topological extension of the rigid rotation $T_{\rho(\tilde{f})}\colon \mathbb{T} \circlearrowleft$, let us start noticing that a continuous map $h\colon \mathbb{T} \circlearrowleft$ with $\deg h = 1$ is always surjective and satisfies $h \circ f = T_\rho \circ h$ if an only if there is a lift $\tilde{h} \circlearrowleft \mathbb{R} \circlearrowleft$ satisfying $\tilde{h} \circ \tilde{f} = \rho(\tilde{f}) + \tilde{h}$. Moreover, we have the following

Exercise 4.12. Show that if $\tilde{f}, \tilde{h}\colon \mathbb{R} \circlearrowleft$ are as above, with $\rho(\tilde{f}) \in \mathbb{R}\backslash\mathbb{Q}$, then $\tilde{h}$ is monotone increasing function. If h is not strictly monotone, i.e. there exists a there is $y \in \mathbb{R}$ such that $\tilde{h}^{-1}(y) = [a, b]$ with $a < b$, then the open interval $\pi(a, b) \subset \mathbb{T}$ is a wandering set for f.

Writing $u := \tilde{h} - id_{\mathbb{R}} \in C^0(\mathbb{T}, \mathbb{R})$, one can easily check that this function must satisfies the following **cohomological equation:**

$$u\big(f(x)\big) - u(x) = \rho(\tilde{f}) - \Delta_{\tilde{f}}(x), \quad \forall x \in \mathbb{T}.$$

In order to prove the existence of such a function u, we need the so called Gottschalk-Hedlund theorem:

Exercise 4.13 (Gottschalk-Hedlund theorem)**.** Let $f\colon M \circlearrowleft$ be a minimal homeomorphisms on the compact metric space M (i.e. every f-orbit is dense in M) $\phi\colon M \to \mathbb{R}$ be a continuous function. Then, there exists a function $u \in C^0(M, \mathbb{R})$ satisffying

$$u \circ f - u = \phi,$$

if and only if there exists $x_0 \in M$ such that

$$\sup_{n\geqslant 1}\left|\sum_{j=0}^{n-1}\phi\big(f^j(x_0)\big)\right| < \infty.$$

Prove this theorem as a consequence of the following steps:

(1) Prove the implication $\Longrightarrow$.
(2) To prove $\Longleftarrow$, let us start considering the map $F\colon M\times\mathbb{R} \circlearrowleft$ given by $F(x,t) := \big(f(x), t+\phi(x)\big)$ for every $(x,t)\in M\times\mathbb{R}$ and show that there is a compact F-invariant set $K \subset \overline{\{F^n(x_0,0) : n\in\mathbb{Z}\}}$ which is minimal for F.
(3) Show that $\mathrm{pr}_1(K) = M$, where $\mathrm{pr}_1\colon M\times\mathbb{R}\to M$ denotes the projection on the first coordinate.
(4) Show that $V_s(K)\cap K = \varnothing$, for every $s\in\mathbb{R}\backslash\{0\}$, where $V_s\colon M\times\mathbb{R}\ni(x,t)\mapsto (x,t+s)\in M\times\mathbb{R}$ denotes the corresponding vertical translation.
(5) Putting together (2), (3) and (4), show that K is the graph of a continuous function, i.e. there is $u\in C^0(M,\mathbb{R})$ such that $K = \{(x,u(x))\in M\times\mathbb{R} : x\in M\}$.
(6) Finally, show that the function u given by (5) satisfies $u\circ f - u = \phi$.

Now, putting together condition (3) of Proposition 4.6 and Gottschalk-Hedlund theorem, we know that there is a continuous function $u\colon K_f\to\mathbb{R}$ such that

$$u\big(f(x)\big) - u(x) = \rho(\tilde f) - \Delta_{\tilde f}(x), \quad \forall x\in K_f.$$

So, defining the map $h\colon K_f\to\mathbb{T}$ by $h(x) := x + u(x)$, for every $x\in K_f$, we get a continuous map h satisffying $h\circ f(x) = T_{\rho(\tilde f)}\circ h(x)$, for every $x\in K_f$.

Observe that if $K_f = \mathbb{T}$ then, f has no wandering points and hence, by Exercise 4.12, h is an orientation preserving homeomorphism and f and $T_{\rho(\tilde f)}$ are indeed topological conjugate.

So, it just remain to consider the case where $K_f\neq\mathbb{T}$. In this case, we still need to define our map h on $\mathbb{T}\backslash K_f$. We will see that this extension of h on K_f can be defined to be constant in each connected component of $\mathbb{T}\backslash K_f$. In fact, if I is any connected component of $\mathbb{T}\backslash K_f$ and its boundary is $\partial I = \{a,b\}$, one has $a,b\in K_f$ and we claim that $h(a) = h(b)$. To prove this, observe that $h\colon K_f\to\mathbb{T}$ is a continuous function defined on the compact set K_f, and thus, it is uniformly continuous. If we have $h(a)\neq h(b)$, we define $\varepsilon := d_{\mathbb{T}}\big(h(a),h(b)\big) > 0$ where $d_{\mathbb{T}}$ is the distance on $\mathbb{T}$ given by

$$d_{\mathbb{T}}(x,y) := \min\{|\tilde x - \tilde y| : \tilde x\in\pi^{-1}(x),\ \tilde y\in\pi^{-1}(y)\}, \quad \forall x,y\in\mathbb{T}.$$

By uniform continuity, there is a $\delta>0$ such that $d_{\mathbb{T}}(h(x),h(y)) < \varepsilon/2$, for every $x,y\in K_f$ with $d_{\mathbb{T}}(x,y)<\delta$. On the other hand, since I is a wandering interval for f, we know that $d_{\mathbb{T}}(f^n(a),f^n(b))\to 0$, as $n\to+\infty$, but since $T_{\rho(\tilde f)}$ is an isometry for the distance $d_{\mathbb{T}}$ we get

$$d_{\mathbb{T}}\Big(h\big(f^n(a)\big), h\big(f^n(b)\big)\Big) = d_{\mathbb{T}}\Big(T^n_{\rho(\tilde f)}\big(h(a)\big), T^n_{\rho(\tilde f)}\big(h(b)\big)\Big) = d_{\mathbb{T}}\big(h(a),h(b)\big) = \varepsilon,$$

for every $n\in\mathbb{Z}$, contradicting the uniform continuity.

So, $h(a) = h(b)$ and we just can define $h\big|_I \equiv h(a) = h(b)$. □

On the circle we have the so called C^∞*-closing lemma:*

Lemma 4.8 (Closing lemma)**.** *Let $f\in\mathrm{Homeo}_+(\mathbb{T})$ be a homeomorphism without periodic orbits. Then, for every non-wandering point $z\in\mathbb{T}$ and any $\varepsilon>0$, there is $t\in(-\varepsilon,\varepsilon)$ such that z is a periodic point for the homeomorphisms $f_t := T_t\circ f\colon \mathbb{T}\circlearrowleft$, where $T_t : x+\mathbb{Z}\mapsto x+t+\mathbb{Z}$.*

Proof. Since $z \in \mathbb{T}$ is non-wandering for a circle homeomorphism, it is a recurrent point, too. So, there exists $n > 0$ such that $d_{\mathbb{T}}(f^n(z), z) < \varepsilon$. Thus, there is $p \in \mathbb{Z}$ such that $\left|\tilde{f}^n(\tilde{z}) - \tilde{z} - p\right| < \varepsilon$, for every $\tilde{z} \in \pi^{-1}(z)$. Without loss of generality, we can assume that $0 \leqslant \tilde{f}^n(\tilde{z}) - \tilde{z} - p < \varepsilon$. Then observe that, since $\tilde{f}$ and $\tilde{T}_t : \mathbb{R} \ni \tilde{x} + \tilde{x} + t$ are increasing functions, we have $\tilde{f}^n_{-\varepsilon}(\tilde{z}) - \tilde{z} - p < 0 \leqslant \tilde{f}^n_0(\tilde{z}) - \tilde{z} - p$, where $\tilde{f}_t := \tilde{T}_t \circ \tilde{f}$. So, by continuity, there is $t \in [-\varepsilon, 0]$ such that $\tilde{f}^n_t(\tilde{z}) = \tilde{z} + p$, for every $\tilde{z} \in \pi^{-1}(z)$, showing that $z \in \mathrm{Per}(f_t)$. □

On the other hand, every periodic point can be turned into a hyperbolic one with an arbitrary small perturbation. In fact, we have

Definition 4.5 (Hyperbolic periodic point)**.** Given a C^1-diffeomorphism $f \colon \mathbb{T} \circlearrowleft$, a periodic point $x \in \mathbb{T}$ is said to be **hyperbolic** when $|Df^n(x)| \neq 1$, for $n \in \mathbb{N}$ such that $f^n(x) = x$.

Exercise 4.14. If $f \in \mathrm{Diff}^r(\mathbb{T})$ with $r \in \mathbb{N}$ and $z \in \mathbb{T}$ is a periodic point with $f^n(z) = z$, then given any $\varepsilon > 0$ there is $g \in \mathrm{Diff}^r(\mathbb{T})$ such that z is a hyperbolic periodic point of g satisfying $g^n(z) = z$ and

$$d_{C^r}(f, g) := \max_{\tilde{x} \in \mathbb{R}} \max_{0 \leqslant i \leqslant r} \left|D^i \tilde{f}(\tilde{x}) - D\tilde{g}(\tilde{x})\right| < \varepsilon,$$

where $\tilde{f}$ and $\tilde{g}$ are appropriate lifts of f and g, respectively.

Finally, any hyperbolic periodic point admits a *continuation:*

Exercise 4.15 (Continuation of hyperbolic periodic points)**.** Let $f \colon \mathbb{T} \circlearrowleft$ be a C^1-diffeomorphism and $z \in \mathbb{T}$ be a hyperbolic periodic point satisfying $f^n(z) = z$. Then, there are $\varepsilon, r >$ such that for every $g \in B^1_\varepsilon := \{h \in \mathrm{Diff}^1(\mathbb{T}) : d_{C^1}(h, f) < \varepsilon\}$ there is a unique element $z_g \in \mathrm{Fix}(g^n)$ with $d_{\mathbb{T}}(z, z_g) < r$.

(**Hint:** use the implicit function theorem).

Then one can state the main result about C^r-generic dynamics of circle diffeomorphisms:

Theorem 4.9 (Density of structural stability for circle diffeomorphisms)**.** *Given any $r \in \mathbb{N}$, the set of so called* Morse-Smale *C^r-diffeomorphisms given by*

$$\mathrm{MS}^r(\mathbb{T}) := \{f \in \mathrm{Diff}^r(\mathbb{T}) : \mathrm{Per}(f) \neq \varnothing,\ \forall z \in \mathrm{Per}(f),\ z \textit{ is hyperbolic}\}$$

is C^r-dense and C^1-open in $\mathrm{Diff}^r(\mathbb{T})$ and every $f \in \mathrm{MS}^r(\mathbb{T})$ is C^1-structurally stable.

Sketch of the proof of Theorem 4.9. The fact that $\mathrm{MS}^r(\mathbb{T})$ is C^1-open follows from Exercise 4.15.

The density of $\mathrm{MS}^r(\mathbb{T})$ follows from Lemma 4.8, Exercises 4.7, 4.14 and 4.15, and Sard's theorem (see for instance [Mil65, §3]).

The fact that every Morse-Smale diffeomorphism is C^1-structurally stable is a consequence of Exercise 4.2. □

5. Dynamics of linear maps

In order to extend the theory of one-dimensional dynamics we developed in §4 to higher dimensions, we first need to understand the dynamics of linear maps on finite dimensional vector spaces. More generally, it will be also useful to study the dynamics of bounded linear operators on Banach spaces.

5.1. **Linear operators on Banach spaces.** Let $(E, \|\cdot\|)$ be a Banach space over either $\mathbb{C}$ or $\mathbb{R}$. A linear operator $A\colon E \circlearrowleft$ is said to be **bounded** when it is continuous and in such a case it holds

$$\|A\| := \sup\{\|Ax\| : x \in E,\ \|x\| = 1\} < \infty. \tag{5}$$

We shall write $\mathcal{L}(E)$ for the vector space of bounded operators on E, which turns into a Banach algebra when it is endowed with norm given by (5).

When E is a Banach space over the complex field $\mathbb{C}$, we define the **spectrum** of the operator $A \in \mathcal{L}(E)$ as the set

$$\sigma(A) := \{\lambda \in \mathbb{C} : A - \lambda I \text{ is not bijective}\},$$

where $I \in \mathcal{L}(E)$ denotes the identity operator.

When E is a Banach space over the real field $\mathbb{R}$, one can define its **complexification** as the Banach space over $\mathbb{C}$ $(E_{\mathbb{C}}, \|\cdot\|_{\mathbb{C}})$ given by $E_{\mathbb{C}} := E \times E$, $\|(x_1, x_2)\| := \max_i \|x_i\|$, for every $(x_1, x_2) \in E_{\mathbb{C}}$ and the multiplication by a complex number is defined by

$$(a + bi)(x_1, x_2) := (ax_1 - bx_2, ax_2 + bx_1), \quad \forall (x_1, x_2) \in E_{\mathbb{C}},\ \forall a + bi \in \mathbb{C}.$$

We can define $E_r := E \times \{0\}$ and $E_i := \{0\} \times E$ as closed linear subspaces of $E_{\mathbb{C}}$, which are isomorphic to E and such that $E_{\mathbb{C}} = E_r \oplus E_i$. Every operator $A \in \mathcal{L}(E)$ induces a new operator $A_{\mathbb{C}} \in \mathcal{L}(E_{\mathbb{C}})$ given by $A_{\mathbb{C}}(x_1, x_2) := (Ax_1, Ax_2)$. On the other hand, an operator $B \in \mathcal{L}(E_{\mathbb{C}})$ is said to be a **real operator** when $B(E_r) \subset E_r$ and $B(E_i) \subset E_i$. One can show that every operator on $E_{\mathbb{C}}$ which is induced by another one on E is indeed a real operator and reciprocally, every real operator on $E_{\mathbb{C}}$ is induced by one defined on E.

The **spectrum** of an operator $A \in \mathcal{L}(E)$ on the real Banach space E is defined as the spectrum of the induced operator on the complexification of E, i.e. by $\sigma(A) := \sigma(A_{\mathbb{C}})$.

Theorem 5.1 (Spectrum and spectral radius)**.** *If E is a Banach space over either $\mathbb{R}$ or $\mathbb{C}$ and $A \in \mathcal{L}(E)\backslash\{0\}$, then it spectrum $\sigma(A)$ is a compact non-empty subset of $\mathbb{C}$. So, if we define the **spectral radius** of A by*

$$r(A) := \max\{|\lambda| : \lambda \in \sigma(A)\},$$

then it holds

$$r(A) := \lim_{n \to +\infty} \|A^n\|^{1/n} = \inf_{n \geqslant 1} \|A^n\|^{1/n}. \tag{6}$$

On the other hand, if $A \in \mathrm{GL}(E)$, then it holds

$$\sigma(A^{-1}) = \left\{z^{-1} \in \mathbb{C} : z \in \sigma(A)\right\}. \tag{7}$$

Proof. See for instance [Mur90, §1.2]. □

We shall write $\mathrm{GL}(E)$ for the group of invertible bounded linear maps on E, i.e. $\mathrm{GL}(E) := \{A \in \mathcal{L}(E) : 0 \notin \sigma(A)\}$.

Definition 5.1 (Hyperbolic linear automorphisms)**.** Given a Banach space E, n invertible linear automorphism $A \in \mathrm{GL}(E)$ is said to be **hyperbolic** when it holds

$$\sigma(A) \cap \{\lambda \in \mathbb{C} : |\lambda| = 1\} = \varnothing.$$

Since the spectrum $\sigma(A)$ is compact, when A is hyperbolic there is constant $\lambda \in (0, 1)$ such that

$$\sigma(A) \cap \left\{z \in \mathbb{C} : \lambda \leqslant |z| \leqslant \lambda^{-1}\right\} = \varnothing. \tag{8}$$

Any number $\lambda \in (0, 1)$ satisfying condition (8) is called a **hyperbolicity constant** for A.

We will need the following result about spectral decompositions:

Theorem 5.2. *Let E be a Banach space and $A \in \mathcal{L}(E)$ be a bounded operator. Let $K_1, K_2 \subset \mathbb{C}$ be two non-empty compact sets such that $\sigma(A) = K_1 \sqcup K_2$. Then there exists two closed proper subspaces $E_1, E_2 \subset E$ such that $E = E_1 \oplus E_2$, $A(E_i) \subset E_i$ and $\sigma\big(A\big|_{E_i}\big) = K_i$, for $i \in \{1, 2\}$.*

Proof. See for instance [DS88, Chapter VII]. □

As a consequence of Theorem 5.1 we get the following result that characterizes the dynamics of hyperbolic linear maps:

Theorem 5.3 (Hyperbolic linear maps)**.** *Let E be a Banach space and $A \in \mathcal{H}(E)$ be a hyperbolic linear automorphism. Let $E = E^s_A \oplus E^u_A$ be the linear splitting induced by the sets $K^s := \sigma(A) \cap \{z \in \mathbb{C} : |z| < 1\}$ and $K^u := \sigma(A) \cap \{z \in C : |z| > 1\}$ in Theorem 5.2.*

Then, given any hyperbolicity constant $\lambda \in \big(0, 1\big)$ for A (i.e. condition (8) holds), there exist a real constant $C > 0$ such that

$$\|A^n v^s\| \leqslant C\lambda^n \|v^s\| \quad \text{and} \quad \big\|A^{-n} v^u\big\| \leqslant C\lambda^n \|v^u\|,$$

for all $n \in \mathbb{N}$ and every $v^s \in E^s_A$ and $v^u \in E^u_A$.

Moreover, for every $\mu \in \big(A_h, 1\big)$, there exists a norm $|\cdot|$ on E, which is equivalent to $\|\cdot\|$, and such that

$$|Av^s| \leqslant \mu |v^s|, \quad \text{and} \quad \big|A^{-1} v^u\big| \leqslant \mu |v^u|, \tag{9}$$

for every $v^s \in E^s_A$ and every $v^u \in E^u_A$.

*Any norm satisfying condition (9) is an called an **adapted norm** for A.*

Corollary 5.4 (Dynamics of hyperbolic linear maps)**.** *If E is a Banach space and $A \in \mathcal{H}(E)$, and $E = E^s_A \oplus E^u_A$ is the linear splitting given by Theorem 5.3, then it holds*

$$E^s_A = \{x \in E : A^n x \to 0, \text{ as } n \to +\infty\},$$
$$E^u_A = \big\{x \in E : A^{-n} x \to 0, \text{ as } n \to +\infty\big\},$$

and $A^n x \to \infty$, as $n \to \pm\infty$, for every $x \in E \backslash \big(E^s_A \cup E^u_A\big)$.

Proof of Corollary 5.4. It is a straightforward consequence of Theorem 5.3. □

Proof of Theorem 5.3. Since E^s_A and E^u_A are closed subspaces, they are Banach spaces themselves. So, let us write $A_i := A\big|_{E^i_A} : E^i_A \circlearrowleft$, for $i \in \{s, u\}$. Let us fix a real number $\lambda \in \big(A_h, 1\big)$. By equation (6), we know that there is $N \in \mathbb{N}$ such that $\|A^n_s\| < \lambda^n$, and, by (7), $\|A^{-n}_u\| < \lambda^n$, for every $n \geqslant N$.

Let us write

$$C := \lambda^{-N} \max_{1 \leqslant n \leqslant N-1} \big\{\|A^n_s\|, \big\|A^{-n}_u\big\|\big\}.$$

Then, given any $n \in \mathbb{N}$ there exist unique non-negative integers r, q, such that $n = qN + r$ and $0 \leqslant r < N$, and hence,

$$\|A^n_s\| \leqslant \|A^r_s\| \big\|A^N_s\big\|^q \leqslant C\lambda^N \lambda^{Nq} < C\lambda^{Nq+r} = C\lambda^n.$$

Similarly one gets

$$\big\|A^{-n}_u\big\| \leqslant \big\|A^{-r}_s\big\| \big\|A^{-N}_s\big\|^q \leqslant C\lambda^N \lambda^{Nq} < C\lambda^{Nq+r} = C\lambda^n.$$

In order to construct an adapted norm, let us fix a real number $\mu \in \big(A_h, 1\big)$ and let λ be an arbitrary number in the open interval $\big(A_H, \mu\big)$. By the above argument, there is a constant $C > 0$ such that $\|A^n_s\| \leqslant C\lambda^n$ and $\|A^{-n}_u\| \leqslant C\lambda^n$, for every

$n \in \mathbb{N}$. Then we define the new norm $|\cdot|_i$ on the closed subspace E_A^i, for $i \in \{s, u\}$ by

$$|x_s|_s := \sum_{j=0}^{+\infty} \frac{1}{\mu^j} \left\| A^j x_s \right\|, \quad \forall x_s \in E_A^s,$$

and

$$|x_u|_u := \sum_{j=0}^{+\infty} \frac{1}{\mu^j} \left\| A^{-j} x_u \right\|, \quad \forall x_u \in E_A^u.$$

Then, given an arbitrary vector $x \in E$, it can be uniquely written as $x = x_s + x_u$, with $x_i \in E_A^i$, an so we can define

$$|x| := \max\{|x_s|_s, |x_u|_u\}.$$

Exercise 5.1. Show that $|\cdot|_s$, $|\cdot|_u$ and $|\cdot|$ are indeed well define norms, and that $|\cdot|$ and $\|\cdot\|$ are equivalent, i.e. there exists a constant $K > 1$ such that

$$K^{-1} \|x\| \leqslant |x| \leqslant K \|x\|, \quad \forall x \in E.$$

Finally observe that

$$|Ax_s| = |Ax_s|_s = \sum_{j \geqslant 0} \frac{\left\| A^{j+1} x_s \right\|}{\mu^j} = \mu(|x_s|_s - \|x_s\|) \leqslant \mu |x_s|,$$

for every $x_s \in E_A^s$ and

$$\left|A^{-1} x_u\right| = \left|A^{-1} x_u\right|_u = \sum_{j \geqslant 0} \frac{\left\| A^{-(j+1)} x_s \right\|}{\mu^j} = \mu(|x_u|_u - \|x_u\|) \leqslant \mu |x_u|,$$

for every $x_u \in E_A^u$. □

6. Local dynamics at hyperbolic periodic points

All along this section M will denote a compact smooth manifold, and we shall write $\mathrm{Diff}^r(M)$ for the group of C^r-diffeomorphisms of M, for any $r \in \mathbb{N}_0$ and assuming that $\mathrm{Diff}^0(M)$ is the group of homeomorphisms of M. We will always assume $\mathrm{Diff}^r(M)$ endowed with the uniform C^r-topology (see for instance Hirsch [Hir76, Chapter 2]).

Definition 6.1 (Hyperbolic periodic point)**.** Given $f \in \mathrm{Diff}^1(M)$, a fixed point $p = f(p) \in M$ is said to be hyperbolic when $Df_p \colon T_pM \circlearrowleft$ is a hyperbolic linear automorphism, where T_pM denotes the tangent space of M at the point p.

More generally, we say that a periodic point $q = f^n(q) \in M$, with $n \geqslant 1$ is **hyperbolic** when $Df_q^n \colon T_qM \circlearrowleft$ is a hyperbolic linear automorphism.

Exercise 6.1. Let $f \in \mathrm{Diff}^1(M)$ and $p \in M$ be a hyperbolic periodic point for f. Show that $f^i(p)$ is a hyperbolic periodic point of f, for every $i \in \mathbb{Z}$.

The fundamental results about the local dynamics at hyperbolic periodic points are the following:

Theorem 6.1 (Continuation and isolation of hyperbolic periodic points)**.** *Let $f \in \mathrm{Diff}^1(M)$ be given and $p = f^n(p) \in M$ be a hyperbolic periodic point. Then, there exists a neighborhood $\mathcal{U} \subset \mathrm{Diff}^1(M)$ of f (in the C^1-topology), a neighborhood $V \subset M$ of p and a continuous map $\Phi_{f,p} \colon \mathcal{U} \to V$ such that $\Phi_{f,p}(f) = p$ and*

$$\mathrm{Fix}(f^n) \cap V = \left\{\Phi_{f,p}(g)\right\}, \quad \forall g \in \mathcal{U}.$$

Theorem 6.2 (Hartman-Grobman theorem)**.** *Let f and p be as in Theorem 6.1. Then there exists a neighborhood U of $0 \in T_pM$, a neighborhood $V \subset M$ of p and a homeomorphism $h\colon U \to V$ such that*

$$h \circ Df_p^n = f^n \circ h,$$

whenever any of these compositions is well defined.

Theorem 6.3 (Local stability at hyperbolic periodic points)**.** *Let f and p be as in Theorem 6.1. Then, there is a neighborhood $\mathcal{U}$ of f in $\mathrm{Diff}^1(M)$ such that the continuation map $\Phi_{f,p}$ given by Theorem 6.1 is well defined on $\mathcal{U}$, and for every $g \in \mathcal{U}$, there are neighborhoods U_f and U_g of the points p and $\Phi(g)$, respectively, and a homeomorphism $h\colon U_f \to U_g$ such that $h(p) = \Phi_{f,p}(g)$ and $h \circ f = g \circ h$, whenever any of these compositions is well defined.*

In order to state the last main result about local dynamics at hyperbolic fixed points, we need to introduce an auxiliary Riemannian metric $\langle \cdot, \cdot \rangle$ on M and assume that M is connected. This metric naturally induces a distance function d on M given by

$$d(p,q) := \inf \left\{ \int_0^1 \|\gamma'(t)\| \, \mathrm{d}t : \gamma\colon [0,1] \to M \ C^1,\ \gamma(0) = p,\ \gamma(1) = q \right\},$$

for every $p, q \in M$, where $\|v\| := \sqrt{\langle v, v \rangle}$ denotes the induced norm on the tangent bundle TM.

One can show that d is indeed a distance function whose induced topology coincides with the original one of M and it complete whenever M is compact. See for instance Lee [Lee18, Chapter 2].

Then, if we consider M endowed with the distance d, $f\colon M \circlearrowleft$ is a C^1 diffeomorphism and $p \in M$ is a fixed point of f, one can define the so called **global stable and unstable sets** of p by

$$\begin{aligned} W^s(p,f) &:= \left\{ x \in M : d\big(f^n(x), p\big) \to 0, \text{ as } n \to +\infty \right\}, \\ W^u(p,f) &:= W^s(p,f^{-1}) = \left\{ x \in M : d\big(f^{-n}(x), p\big) \to 0, \text{ as } n \to +\infty \right\}, \end{aligned} \tag{10}$$

respectively.

Similarly, give any $\varepsilon > 0$ one defines the **local stable and unstable sets** of p by

$$\begin{aligned} W^s_\varepsilon(p,f) &:= \left\{ x \in M : d\big(f^n(x), p\big) < \varepsilon,\ \forall n \geqslant 0 \right\}, \\ W^u_\varepsilon(p,f) &:= W^s_\varepsilon(p,f^{-1}) = \left\{ x \in M : d\big(f^{-n}(x), p\big) < \varepsilon,\ \forall n \geqslant 0 \right\}. \end{aligned} \tag{11}$$

Theorem 6.4 (Local stable/unstable manifold theorem)**.** *Let M be a compact smooth manifold, $f \in \mathrm{Diff}^r(M)$ with $r \geqslant 1$ and $p \in M$ be a hyperbolic fixed point for f with index $s := \dim E^s_p$. Then, there exists $\varepsilon > 0$ such that $W^s_\varepsilon(p,f) \subset W^s(p,f)$, and there is a C^r-embedding $\Phi^s_{p,\varepsilon}\colon \mathbb{D}^s := \{x \in \mathbb{R}^s : \|x\| < 1\} \to M$ such that $\Phi^\sigma_{p,\varepsilon}(0) = p$, $D\Phi^\sigma_{p,\varepsilon}(0)(\mathbb{R}^s) = E^s_p$ and $\Phi^s_{p,\varepsilon}(\mathbb{D}^s) = W^s_\varepsilon(p,f)$.*

An analogous result holds for the unstable manifold.

Corollary 6.5 (Global stable/unstable manifold)**.** *If M, f and p are as in Theorem 6.4, then there exists an injective C^r-immersion $\Phi^s_p\colon \mathbb{R}^s \to M$ such that $\Phi^s_p(0) = p$, $D\Phi^s_p(\mathbb{R}^s) = E^s_p$ and $\Phi^s_p(\mathbb{R}^s) = W^s(p,f)$.*

An analogous result holds for the unstable manifold.

Let us start sketching the main ideas needed to prove these results.

Proof of Theorem 6.1. This result is a consequence of (the proof of) the implicit function theorem and the fact that $1 \notin \sigma(Df^n_p)$ when p is a hyperbolic periodic point. □

We will postpone the proof of Theorems 6.2, 6.3 and 6.4 for the next section.

6.0.1. *Local dynamics on Banach spaces.* Let $(E, \|\cdot\|)$ be a Banach space over $\mathbb{R}$. We shall write $C_b^0(E)$ for the space of bounded continuous functions from E to itself. This is Banach space when it is endowed with the norm

$$\|\phi\| := \sup_{x\in E} \|\phi(x)\|, \quad \forall \phi \in C_b^0(E).$$

Our first result is the following

Proposition 6.6. *Let $A\colon E \circlearrowleft$ be a linear automorphism and $\phi \in C_b^0(E)$ be a Lipschitz function such that*

$$\operatorname{Lip}\phi := \sup_{x\neq y} \frac{\|\phi(x)-\phi(y)\|}{\|x-y\|} < \left\|A^{-1}\right\|^{-1}.$$

Then, $A+\phi$ is Lipschitz homeomorphism.

Proof. In order to show that $A+\phi$ is injective, it is enough to notice that

$$\begin{aligned}\|(A+\phi)(x)-(A+\phi)(y)\| &\geqslant \|Ax-Ay\| - \|\phi(x)-\phi(y)\| \\ &\geqslant \left\|A^{-1}\right\|^{-1}\|x-y\| - \operatorname{Lip}\phi\,\|x-y\| \\ &\geqslant \left(\left\|A^{-1}\right\|^{-1} - \operatorname{Lip}\phi\right)\|x-y\|,\end{aligned}$$

for all $x, y \in E$.

To prove that $A+\phi$ is surjective let us consider an arbitrary $y \in E$ and notice that there exists $x \in E$ satisffying $(A+\phi)x = y$ if and only if

$$x = A^{-1}\big(y-\phi(x)\big).$$

So, if we define the operator $\Xi_y\colon E \to E$ by

$$\Xi_y(z) := A^{-1}\big(y-\phi(z)\big), \quad \forall z \in E,$$

we observe that

$$\left\|\Xi_y(z)-\Xi_y(z')\right\| = \left\|A^{-1}\phi(z')-A^{-1}\phi(z)\right\| \leqslant \left\|A^{-1}\right\|\operatorname{Lip}\phi\,\|z-z'\|,$$

for every $z, z' \in E$. So, Ξ_y is a contraction and by Theorem 2.1, Ξ_y has a fixed point. □

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